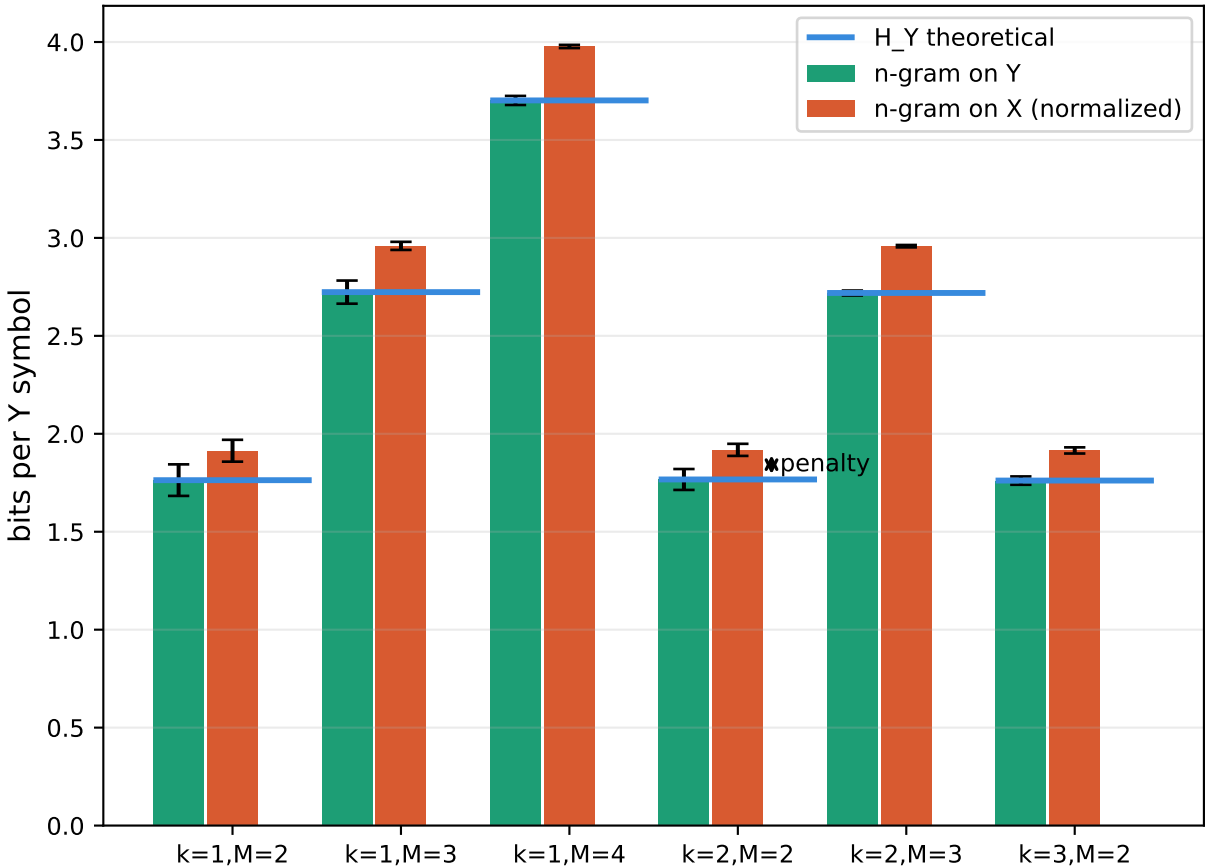
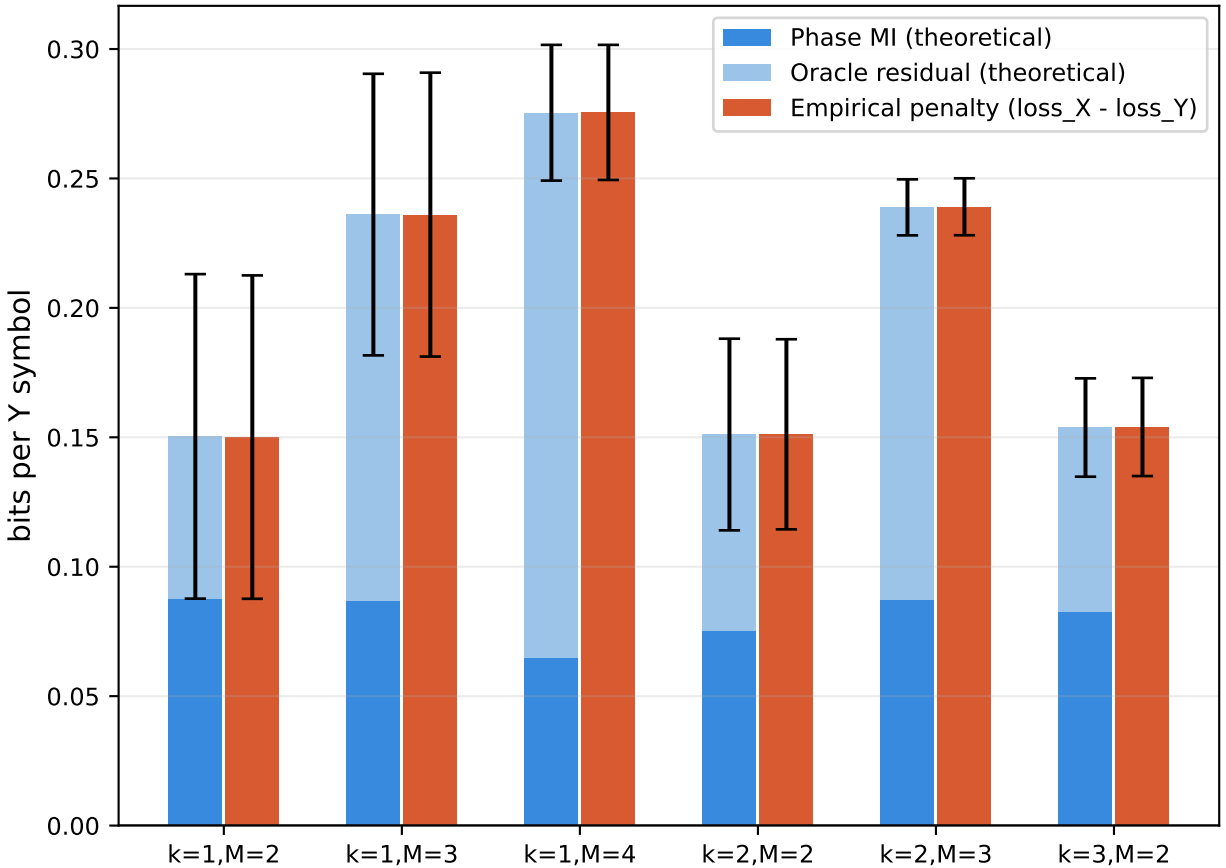

(a) Per-symbol loss comparison across configurations
H_Y theoretical
n-gram on Y
n-gram on X (normalized)
penalty
bits per Y symbol
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
k=1,M=2
k=1,M=3
k=1,M=4
k=2,M=2
k=2,M=3
k=3,M=2


(b) Fragmentation penalty decomposition: theoretical vs empirical
Phase MI (theoretical)
Oracle residual (theoretical)
Empirical penalty (loss_X - loss_Y)
bits per Y symbol
0.00
0.05
0.10
0.15
0.20
0.25
0.30
k=1,M=2
k=1,M=3
k=1,M=4
k=2,M=2
k=2,M=3
k=3,M=2